Start-State Critic Bias: A Per-Update Diagnostic for Actor–Critic Deep RL

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Abstract

Actor–critic training monitors and optimizes critic error on the rollout distribution, but policy improvement strictly depends on critic accuracy at the start-state distribution. We demonstrate that this distribution mismatch makes the absolute Bellman residual (ϵ_u) the wrong quantity for certifying update health. Theoretically, we show that the relevant criterion is a dimensionless ratio, and that any fixed absolute tolerance on ϵ_u eventually becomes vacuous as the policy improves. The mechanical consequence of this mismatch is revealed by an exact policy-improvement identity: the operative quantity is the **start-state critic bias** $\hat{\Delta}_k = J(\pi_k) - \mathbb{E}_\nu V_{k+1}$. Empirically, we audit over 86,000 updates across PPO, SAC, and TRPO. Because it measures the right error on the wrong distribution, ϵ_u is near-chance at predicting policy harm (pooled median AUROC 0.46). In contrast, the start-state bias reflects the true covariate shift and outperforms ϵ_u on 234 of 236 paired seeds. We confirm this is a persistent structural boundary, rather than an artifact of baseline loss design: directly targeting the bias via a ν-weighted critic loss does not remove the retrospective boundary (lag-1 AUROC remains at chance). We conclude that the standard Bellman residual cannot reliably certify update health; practitioner monitoring must shift toward the start-state bias.

1 Introduction

Actor–critic algorithms in deep reinforcement learning (RL)—including PPO [Schulman et al., 2017], SAC [Haarnoja et al., 2018], and TRPO [Schulman et al., 2015a]—update a policy iteratively against a learned value function. The standard analytic guarantee for the success of these updates typically relies on a bound of the form:

$$J(\pi_{k+1}) - J(\pi_k) \;\geq\; A_k \;-\; c \cdot \epsilon_u, \tag{1}$$

where ϵ_u is a norm of the Bellman residual $T^\pi V - V$ and A_k is the advantage signal. In practice, the scalar ϵ_u is the value-function loss minimized by nearly every deep actor–critic implementation [Huang et al., 2022, Andrychowicz et al., 2021], and is the primary quantity practitioners monitor to assess whether their critic is "good enough" to support policy improvement.

In this paper, we show that this reliance on the absolute Bellman residual is structurally flawed. The underlying issue is a distribution mismatch: actor–critic algorithms measure and optimize critic error on the discounted rollout distribution (d^{π_k}), but true policy improvement depends strictly on the critic's accuracy at the start-state distribution (ν). We expose the mechanics of this mismatch and demonstrate that it makes ϵ_u the wrong object for monitoring update health.

Mechanism: the start-state bias as covariate shift. The failure of the Bellman residual to track improvement is mechanically explained by an exact algebraic identity (Proposition 1):

$$J(\pi_{k+1}) - J(\pi_k) \;=\; A_k \;-\; \hat{\Delta}_k, \qquad \hat{\Delta}_k = J(\pi_k) - \mathbb{E}_{s\sim\nu}[V_{k+1}(s)]. \tag{2}$$

We call $\hat{\Delta}_k$ the **start-state critic bias**: it is the post-update critic's error *at the initial-state distribution* ν. The Bellman residual ϵ_u measures critic error on the rollout distribution d^{π_k}; it is silent about accuracy at ν. The two distributions disagree even on-policy because d^{π_k} is the discounted state-visitation distribution and ν is its starting point. Because it measures the right kind of error on the wrong distribution, ϵ_u cannot reliably certify improvement.

Theory: absolute tolerances eventually become vacuous. We first give a constructive argument for why the distribution mismatch breaks standard bounds. We show that the relevant theoretical tolerance is the *dimensionless ratio* ϵ_u/A_k. As training progresses and the policy approaches a local optimum, the advantage signal A_k naturally tends toward zero. Consequently, any fixed absolute tolerance ϵ^{fixed} eventually becomes too large relative to the available improvement signal, rendering the improvement guarantee (1) vacuous. We formalize this in Proposition 2, showing that fixed-tolerance critics eventually fail to certify even genuine improvements.

Empirical structural limit across PPO, SAC, and TRPO. We validate this structural limit with an audit of over 86,000 actor–critic updates across eight environments and 237 seeds. As predicted by the distribution mismatch, the canonical Bellman residual is near-chance at predicting whether an update will be beneficial or harmful (pooled median AUROC 0.46). In contrast, the start-state bias $-\hat{\Delta}_k$ reflects the true covariate shift and outperforms ϵ_u on 234 of 236 paired seeds. This outperformance holds uniformly across PPO, SAC, and TRPO, and is robust to a $\sim$200-cell hyperparameter factorial. We confirm the signal reflects true critic generalization by replicating the diagnostic on fully independent held-out rollouts (AUROC 0.61).

The retrospective boundary is persistent. A natural intervention is to simply target the bias directly in the critic loss. We evaluate a ν-weighted critic loss and find it fails to remove the boundary: the lag-1 prospective AUROC remains at chance. The diagnostic uses the post-update critic V_{k+1}, making it intrinsically retrospective; our null result demonstrates this boundary is persistent due to the optimization order, not merely a deficiency of standard MSE loss. This reveals a limit on online actor-critic control: practitioners can use start-state bias for retrospective auditing and checkpoint rollback, but not as a one-step-ahead prospective controller.

Contributions. Our contributions are: (i) identification of the distribution mismatch between d^{π_k} and ν as the mechanism decoupling value loss from policy improvement; (ii) the exact identity $J(\pi_{k+1}) - J(\pi_k) = A_k - \hat{\Delta}_k$, isolating the start-state bias; (iii) a no-go theorem (Proposition 2) for absolute Bellman residuals; and (iv) a large-scale empirical audit proving that practitioner monitoring must shift from rollout-based TD errors to relative, start-state-aware quantities.

2 Related work

Distribution shift in RL. Prior work on distribution shift has largely focused on the behavior/target policy mismatch in off-policy learning [Kakade and Langford, 2002, Islam et al., 2019]. We identify the mismatch between the rollout distribution d^{π_k} and the start-state distribution ν as the operative gap even in strictly on-policy methods like PPO. This reframes critic monitoring as a fundamental covariate shift problem inherent to the actor–critic objective itself.

Empirical studies of critic failure. Our work provides a structural lens for the critical evaluation of deep RL implementations [Henderson et al., 2018, Engstrom et al., 2020]. Ilyas et al. [2020] established that critics fail to fit the true value function globally—the network solves the supervised task but not the estimation problem. The distribution mismatch explains *why* this is catastrophic for monitoring: even accepting the supervised rollout quality produced by a run, the scalar used to measure that quality (ϵ_u) is disconnected from the distribution that drives policy improvement (ν).

Value overestimation. The overestimation literature [Fujimoto et al., 2018] motivates critic design choices, such as clipped double-Q in SAC, by targeting overestimation on the rollout distribution. The covariate shift implies a sharper target: what matters for policy improvement is the bias at ν, not the average error over d^{π_k}. Our SAC audit confirms that while these methods stabilize training on the rollout, they do not close the structural gap between ϵ_u and $\hat{\Delta}_k$ as diagnostic predictors.

3 Audit setup

We audit four actor–critic configurations spanning three algorithm families.

Algorithms and environments. PPO [Schulman et al., 2017] on the canonical 8-environment suite at 200k–500k timesteps each: discrete-action LunarLander-v3, CartPole-v1, Acrobot-v1, and MuJoCo continuous control on Hopper-v5, HalfCheetah-v5, Walker2d-v5, Ant-v5, Humanoid-v5; 20 seeds per environment (160 runs). PPO-MSE: an ablation that replaces PPO's clipped value loss with a pure mean-squared TD error, on the same 8 envs $\times$ 5 seeds. SAC [Haarnoja et al., 2018] on the three MuJoCo envs where it is known to train (≥ 10 seeds each). TRPO [Schulman et al., 2015a] on LunarLander-v3 and Hopper-v5 (≥ 9 seeds each). Three seeds stalled below 10 updates (one PPO CartPole seed plateaued at the 500-step solved threshold, one SAC Hopper, one TRPO LunarLander) and are excluded from per-seed AUROC; 237 seeds remain (Table 1, n column). Total: $\sim$86,000 actor–critic updates with paired (diagnostic, ground-truth ΔJ_k) records.

For each update we log $\widehat{A}_k$, $\widehat{\hat{\Delta}}_k$ (the start-state estimator, eq. (6)), $\widehat{\epsilon}_u$ (mean squared TD error on the rollout, post-critic-update, the canonical scalar minimized in PPO/SAC/TRPO codebases), and the ground-truth return $\widehat{J}(\pi_k)$. The harmful label is $1[\widehat{\Delta J}_k < 0]$, where $\widehat{\Delta J}_k = \widehat{J}(\pi_{k+1}) - \widehat{J}(\pi_k)$ from successive rollout batches. The harmful label is undefined on the final update of each run.

Hyperparameter factorials. To rule out tuning artifacts we run three factorial sweeps. LunarLander: 4×4 over value-epochs $\in \{5, 10, 20, 40\}$ and clip-$\epsilon \in \{0.1, 0.2, 0.3, 0.4\}$, 5 seeds per cell (80 runs). HalfCheetah and Hopper: 3×2 over value-epochs $\in \{5, 10, 20\}\times$ clip-$\epsilon \in \{0.1, 0.2\}$ plus a 3-cell value-learning-rate sweep at the default cell, 5 seeds each (45 runs per env). All factorials use the same logging schema. Full details in Section G.

4 The central lens: an exact identity at the start-state distribution

We consider an infinite-horizon discounted MDP $(\mathcal{S}, \mathcal{A}, P, R, \gamma, \nu)$ with state space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel $P : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$, reward R, discount $\gamma \in [0, 1)$, and initial-state distribution $\nu \in \Delta(\mathcal{S})$. For a policy π, write V^π for its value function, $J(\pi) = \mathbb{E}_{s_0 \sim \nu}[V^\pi(s_0)]$ for its expected return, T^π for the Bellman operator, and d^π for the discounted state-visitation distribution from ν:

$$(T^\pi V)(s) = \mathbb{E}_{a\sim\pi}\big[R(s,a) + \gamma \mathbb{E}_{s'}[V(s')]\big], \qquad (d^\pi)^\top = (1-\gamma)\,\nu^\top (I - \gamma P^\pi)^{-1}.$$

Consider an iterative algorithm that, at iteration k, holds π_k and a critic estimate V_k, and produces (π_{k+1}, V_{k+1}). The estimate V_{k+1} need not coincide with $V^{\pi_{k+1}}$.

Proposition 1 (Improvement identity at the initial-state distribution)**.** *For any policies π_k, π_{k+1} and any value function $V_{k+1} : \mathcal{S} \to \mathbb{R}$,*

$$J(\pi_{k+1}) - J(\pi_k) \;=\; A_k \;-\; \hat{\Delta}_k, \tag{3}$$

where

$$A_k := \tfrac{1}{1-\gamma}\, \mathbb{E}_{s\sim d^{\pi_{k+1}}}[(T^{\pi_{k+1}} V_{k+1})(s) - V_{k+1}(s)] \text{ and } \hat{\Delta}_k := J(\pi_k) - \mathbb{E}_{s\sim\nu}[V_{k+1}(s)]. \tag{4}$$

Proof. The Bellman residual at π_{k+1} on V_{k+1} is $T^{\pi_{k+1}} V_{k+1} - V_{k+1} = r^{\pi_{k+1}} + \gamma P^{\pi_{k+1}} V_{k+1} - V_{k+1}$. Substituting $(d^{\pi_{k+1}})^\top = (1-\gamma)\nu^\top (I - \gamma P^{\pi_{k+1}})^{-1}$ gives

$$(1-\gamma) A_k = \nu^\top (I - \gamma P^{\pi_{k+1}})^{-1} \big(r^{\pi_{k+1}} - (I - \gamma P^{\pi_{k+1}}) V_{k+1}\big) = J(\pi_{k+1}) - \mathbb{E}_\nu V_{k+1}.$$

Subtracting $\hat{\Delta}_k = J(\pi_k) - \mathbb{E}_\nu V_{k+1}$ gives $A_k - \hat{\Delta}_k = J(\pi_{k+1}) - J(\pi_k)$. □

Definition 1. *The* certification gap *at iteration k is* $\mathrm{cg}_k := A_k - \hat{\Delta}_k$.

By Proposition 1, $\mathrm{cg}_k > 0 \iff J(\pi_{k+1}) > J(\pi_k)$. The identity is exact for any V_{k+1}, and we verify it holds to machine precision in tabular RandomMDPs (Section B, median residual 5×10^{-16}, max 4×10^{-13}).

Decomposition vs. forecasting. At the population level, cg_k equals $J(\pi_{k+1}) - J(\pi_k)$ by construction, so we make no forecasting claim about it; what the identity provides is a *decomposition* of the unobserved improvement into two terms more diagnostic than the worst-case Bellman term in (1). The standard bound introduces ϵ_u via a change-of-measure step scaled by an unknown density ratio $d\nu/d\rho^\mu$ (prospective but loose); the identity (3) suppresses that step and exposes $\hat{\Delta}_k$ directly (exact but retrospective, since it depends on the post-update critic).

Practical estimators. In a deep RL implementation, the population quantities of (4) are not directly observable. The estimators we use are:

$$\widehat{A}_k \;:=\; \mathbb{E}_{(s,a)\sim B_k}\left[\tfrac{\pi_{k+1}(a|s)}{\pi_k(a|s)}\,\hat{A}^{\mathrm{GAE}}_{V_k}(s,a)\right], \tag{5}$$

$$\widehat{\hat{\Delta}}_k \;:=\; \widehat{J}(\pi_k) \;-\; \tfrac{1}{|\mathcal{S}_0|}\sum_{s_0\in\mathcal{S}_0} V_{k+1}(s_0), \tag{6}$$

$$\widehat{\hat{\Delta}}_k^{\,\mathrm{res}} \;:=\; \frac{1}{1-\gamma}\,\mathbb{E}_{(s,a,s')\sim B_k}\big[r(s,a) + \gamma V_{k+1}(s') - V_{k+1}(s)\big]\,. \tag{7}$$

Here B_k is the on-policy rollout batch under π_k, $\hat{A}^{\,\mathrm{GAE}}$ is the GAE [Schulman et al., 2015b] advantage at the pre-update critic V_k, $\widehat{J}(\pi_k)$ is the mean episode return on the rollout, and $\mathcal{S}_0$ is the set of start states sampled in the rollout. The empirical certification gap is $\widehat{\mathrm{cg}}_k := \widehat{A}_k - \widehat{\hat{\Delta}}_k$. Three sources of approximation error separate $\widehat{\mathrm{cg}}_k$ from the population cg_k: (i) the importance-weighted advantage uses V_k rather than V_{k+1}; (ii) the rollout distribution is d^{π_k} rather than $d^{\pi_{k+1}}$; (iii) finite-sample noise in $\widehat{J}(\pi_k)$ and the start-state empirical mean. The empirical $\widehat{\mathrm{cg}}_k$ is therefore a noisy and biased estimator of cg_k. We adopt the start-state form (6) for $\widehat{\hat{\Delta}}_k$ throughout; the alternative residual-based estimator (7) has lower per-update variance but agrees with the start-state estimator only in expectation; we discuss the comparison in Section D.

5 The failure of the absolute Bellman residual

Because of the distribution mismatch between $d^{\pi_{k+1}}$ and ν, we show that a fixed absolute tolerance on ϵ_u is the wrong theoretical target. Consider the mixture-policy update used in CPI [Kakade and Langford, 2002] and TRPO [Schulman et al., 2015a]: $\pi^\alpha_{k+1} = \alpha\bar{\pi}_k + (1-\alpha)\pi_k$ with $\bar{\pi}_k$ greedy on V_{k+1} and $\alpha \in [0,1]$. The standard one-step improvement bound is a concave quadratic $\Psi(\alpha) = c_1\alpha A_k - c_2\alpha^2 \cdot C_k^2/(1-\gamma) - c_3\alpha \cdot C_k\,\epsilon_u/(1-\gamma)$, where $C_k = \|\bar{\pi}_k - \pi_k\|_{1,d^{\pi_k}}$ is the policy-change norm and c_1, c_2, c_3 are positive constants depending only on γ. Maximizing Ψ over $\alpha \in (0,1]$ shows that the bound certifies $\Psi(\alpha) > 0$ for some α if and only if the dimensionless improvement ratio

$$\tilde{\rho}_k \;:=\; \frac{2(1-\gamma)A_k}{C_k\,\epsilon_u}$$

exceeds one. The relevant tolerance is the ratio ϵ_u/A_k, not ϵ_u in absolute terms.

Proposition 2 (Failure of fixed absolute tolerance)**.** *For any fixed $\epsilon^{\mathit{fixed}} > 0$ and any $\gamma \in (0,1)$, there exist MDPs and convergent policy sequences (π_k) for which every critic satisfying the RPI invariant $T^{\pi_k}V_{k+1} \geq V_{k+1}$ with $\epsilon_{u,k} = \epsilon^{\mathit{fixed}}$ yields $\tilde{\rho}_k \leq 1$ for all k.*

Proposition 2 is proved in Section C by a two-state construction; we read it as an *existence* result, demonstrating that the standard improvement bound can be made vacuous by adversarially chosen MDPs and policy sequences rather than as a quantitative law over all MDPs. The mechanism is qualitative but clear: as $\pi_k \to \pi^*$, the optimization signal $A_k \to 0$, and any fixed absolute tolerance eventually exceeds $C_k\epsilon^{\mathrm{fixed}}/[2(1-\gamma)]$. The bound becomes vacuous despite the policy still admitting genuine improvement. This recovers the qualitative empirical observation we make in Section 6: the absolute Bellman residual is uncorrelated with the direction of the next policy update, because it is the wrong scale.

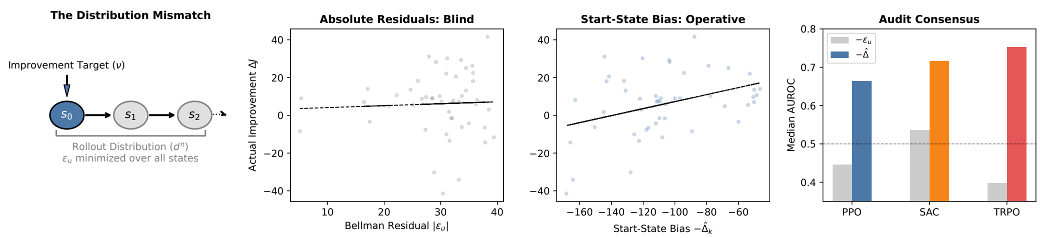

Figure 1: **The distribution mismatch decouples value-loss from improvement.** (Panel 1) The Bellman residual ϵ_u is minimized over the rollout distribution d^π (all states traversed in an episode). However, the true objective—policy improvement—depends exclusively on critic accuracy at the start states ν. (Panel 2) Because it measures the wrong distribution, the absolute Bellman residual $|\epsilon_u|$ is near-blind to ground-truth improvement ΔJ_k (Humanoid-v5, representative run). (Panel 3) In contrast, measuring the critic's error exactly at the start states (the start-state bias $-\hat{\Delta}_k$) exposes the operative relationship. (Panel 4) Median per-seed AUROC for harm prediction: $-\epsilon_u$ (gray) sits at or below chance across PPO, SAC, and TRPO; $-\hat{\Delta}_k$ (colored) is reliably predictive in every algorithm. Full per-environment table is Table 1.

6 Results

Figure 1 previews the headline pattern; Table 1 reports the full per-environment median AUROC for $-\widehat{\epsilon}_u$ and $-\widehat{\hat{\Delta}}_k$ across all four configurations. The pattern is uniform: $\widehat{\epsilon}_u$ pools to 0.46 AUROC across 237 seeds while $-\widehat{\hat{\Delta}}_k$ pools to 0.68. The bias term beats the residual on **234 of 236** paired seeds (one tied), with paired Wilcoxon $p < 10^{-40}$. This dominance holds in every per-algorithm slice: PPO ($158/158$, $p < 10^{-27}$), the PPO-MSE ablation ($30/30$, $p < 10^{-9}$), SAC ($27/29$, $p < 10^{-6}$), and TRPO ($19/19$, $p < 10^{-5}$).

The PPO-MSE ablation rules out clipping artifacts. A reasonable concern is that PPO's value loss is *clipped* (the $\widehat{\epsilon}_u$ a practitioner sees is the clipped MSE, not the raw MSE). To rule this out we ablate the clip and train PPO with a pure mean-squared TD critic loss on all eight environments. Table 1 (PPO-MSE rows) shows the same pattern: $-\widehat{\epsilon}_u$ at chance (0.50 pooled), $-\widehat{\hat{\Delta}}_k$ at 0.67. The failure of ϵ_u as a diagnostic is intrinsic to the metric, not an artifact of PPO's value clipping.

The decomposition isolates $\hat{\Delta}_k$. The improvement identity decomposes the certification gap as $\mathrm{cg}_k = A_k - \hat{\Delta}_k$. Auditing the components individually (Section F) reveals that the predictive signal lives almost entirely in $\hat{\Delta}_k$: across the 8 PPO envs the $-\hat{\Delta}_k$-only AUROC matches the full cg_k AUROC to within ± 0.01, while A_k-alone is pooled 0.39 (*below chance*). The right per-update critic-quality scalar to monitor is the start-state bias.

The A_k paradox. The surrogate advantage A_k pools to AUROC 0.39—*below chance*—as a predictor of ΔJ_k. This is striking on its face: A_k is the very quantity actor–critic methods are derived to maximize. The identity (3) makes the mechanism mechanical: $\Delta J_k = A_k - \hat{\Delta}_k$, so $\mathrm{Corr}(A_k, \Delta J_k)$ flips sign once $\mathrm{Cov}(A_k, \hat{\Delta}_k) > \mathrm{Var}(A_k)$.

Table 2 shows the coupling pattern. It is strongly positive on the high-dimensional MuJoCo tasks where each rollout contains few complete episodes—Humanoid ($\rho = 0.57$), HalfCheetah (0.66), Ant (0.49)—and negative on the episode-rich Hopper (-0.42) and Walker2d (-0.23). We read this as overfitting on the training rollout: when A_k is large, the policy has moved toward actions whose pre-update GAE (computed against the rollout-fit critic V_k) was favourable, and the post-update critic V_{k+1} is then over-confident at the same start states, inflating $\hat{\Delta}_k$. The diagnostic recommendation is unchanged either way: $\hat{\Delta}_k$ outperforms ϵ_u as a predictor in every environment, regardless of the sign of its coupling with A_k.

Table 1: **Per-environment median harm-prediction AUROC, by algorithm.** For each (algorithm, environment) cell we compute the per-seed AUROC of $-\widehat{\varepsilon}_u$ and $-\widehat{\hat{\Delta}}_k$ for predicting $\Delta J_k < 0$ on that seed; the table reports seed medians. '$\hat{\Delta}$ wins' is the per-seed paired count favoring $-\hat{\Delta}_k$; p is the one-sided paired Wilcoxon ($-\hat{\Delta}_k > -\varepsilon_u$). 'base' is the harmful-update fraction (chance AUROC is 0.5; chance AUPRC equals the base rate). Cells with $n < 5$ seeds report '—' for the Wilcoxon p.

Algorithm	Environment	n	$-\varepsilon_u$	$-\hat{\Delta}_k$	$\hat{\Delta}$ wins	p	base
PPO	LunarLander	20	0.45	**0.72**	20/20	1e-06	0.48
PPO	CartPole	19	0.38	**0.61**	19/19	2e-06	0.29
PPO	Acrobot	20	0.32	**0.81**	19/19	2e-06	0.42
PPO	Hopper	20	0.46	**0.66**	20/20	1e-06	0.31
PPO	HalfCheetah	20	0.55	**0.63**	20/20	1e-06	0.44
PPO	Walker2d	20	0.45	**0.62**	20/20	1e-06	0.41
PPO	Ant	20	0.43	**0.67**	20/20	1e-06	0.48
PPO	Humanoid	20	0.52	**0.75**	20/20	1e-06	0.46
PPO-MSE	LunarLander	5	0.50	**0.72**	5/5	0.031	0.48
PPO-MSE	Hopper	5	0.51	**0.73**	5/5	0.031	0.31
PPO-MSE	HalfCheetah	5	0.58	**0.63**	5/5	0.031	0.44
PPO-MSE	Walker2d	5	0.43	**0.62**	5/5	0.031	0.41
PPO-MSE	Ant	5	0.45	**0.59**	5/5	0.031	0.49
PPO-MSE	Humanoid	5	0.53	**0.75**	5/5	0.031	0.47
PPO-NU	Hopper	5	0.39	**0.65**	5/5	0.031	0.33
PPO-NU	HalfCheetah	5	0.54	**0.58**	5/5	0.031	0.42
PPO-NU	Humanoid	5	0.52	**0.76**	5/5	0.031	0.47
SAC	Hopper	9	0.56	**0.72**	8/9	0.027	0.40
SAC	HalfCheetah	10	0.51	**0.56**	9/10	0.003	0.36
SAC	Walker2d	10	0.54	**0.78**	10/10	0.001	0.45
TRPO	LunarLander	9	0.39	**0.79**	9/9	0.002	0.44
TRPO	Hopper	10	0.39	**0.71**	10/10	0.001	0.33
pooled	*PPO*	159	0.45	**0.68**	158/158	6e-28	—
pooled	*PPO-MSE*	30	0.50	**0.67**	30/30	9e-10	—
pooled	*PPO-NU*	15	0.52	**0.65**	15/15	3e-05	—
pooled	*SAC*	29	0.53	**0.68**	27/29	7e-07	—
pooled	*TRPO*	19	0.39	**0.74**	19/19	2e-06	—
ALL	**(4 algos, 19 cells)**	**252**	**0.46**	**0.68**	**249/251**	**9e-43**	—

Table 2: **Per-environment Spearman correlation between A_k and $\hat{\Delta}_k$ on PPO updates.** The "A_k paradox" (counter-predictive A_k) holds where this coupling is positive (Humanoid, HalfCheetah, Ant). The dominance of $-\hat{\Delta}_k$ over $-\epsilon_u$ in Table 1 holds regardless.

Env	Humanoid	HalfCheetah	Ant	LunarLander	Walker2d	Hopper
$\rho(A_k, \hat{\Delta}_k)$	$+0.57$	$+0.66$	$+0.49$	$+0.22$	-0.23	-0.42
n updates	4,704	2,940	2,940	2,940	2,940	2,842

7 Empirical limits of prospective control

The distribution mismatch forces a strict boundary on how the diagnostic can be used: it is a powerful retrospective auditor, but it cannot be easily converted into a prospective online controller.

The retrospective boundary: lag-1 collapse. Because the diagnostic $\hat{\Delta}_k$ is computed using the post-update critic V_{k+1}, it is intrinsically retrospective. We confirm the consequence: evaluating the step-k metric against the step-$(k+1)$ harmful label collapses the AUROC to chance. Pooled lag-1

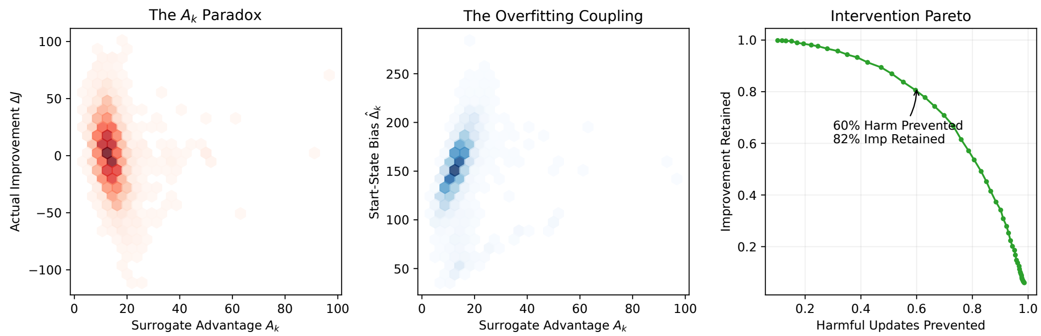

Figure 2: **The A_k paradox and a checkpoint-rollback gate.** (Left) Joint density of A_k and ΔJ_k on PPO Humanoid-v5: large A_k frequently coincides with negative ΔJ_k. (Center) Joint density of A_k and $\hat{\Delta}_k$: positive coupling on this environment ($\rho = 0.57$). (Right) Trade-off curve for the post-hoc gate of Section 7. Sweeping the threshold τ on $\hat{\Delta}_k$ retrospectively, ∼60% of harmful updates can be flagged while retaining ∼80% of the cumulative improvement on Humanoid-v5.

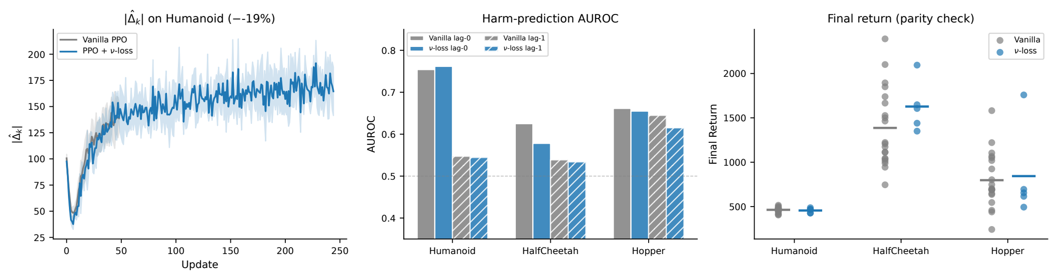

Figure 3: **Targeting the bias directly.** Training with a ν-weighted critic loss ($\lambda = 1.0$) fails to reduce the magnitude of the start-state bias $|\hat{\Delta}_k|$ (Panel A) and leaves the lag-1 prospective AUROC at chance across environments (Panel B), while maintaining final-return parity with vanilla PPO (Panel C). The diagnostic remains retrospective even when targeted by the loss.

AUROC across the PPO grid is 0.51, with mild heterogeneity (Hopper 0.65, Walker2d 0.57, others within ± 0.05 of 0.50). The signal is real and concurrent, but it does not look one step ahead.

Targeting the bias directly: a structural limit. If the boundary is merely an artifact of missing loss terms, targeting the bias directly in the critic loss should allow the critic to generalize prospectively. We evaluate a ν-weighted critic loss (adding a start-state MSE penalty $L = \text{MSE}(V(s), G_s) + \lambda \cdot \text{MSE}(V(s_0), G_0)$) on Humanoid-v5, HalfCheetah-v5, and Hopper-v5 with $\lambda = 1.0$. Figure 3 shows this addition fails to drive down $|\hat{\Delta}_k|$ compared to vanilla PPO and leaves the lag-1 AUROC at chance (pooled 0.54 vs vanilla 0.56), while maintaining final-return parity. The diagnostic remains stubbornly retrospective even when explicitly targeted by the loss. We interpret this null result as a structural observation: the retrospective boundary is a persistent consequence of the optimization order, not merely a deficiency of the standard MSE loss.

Operational feasibility: post-hoc gate via checkpoint and rollback. While standard optimization cannot prospectively fix the bias, practitioners can still embrace its retrospective nature operationally. We evaluate a simple checkpoint-and-rollback pattern: compute V_{k+1} on a candidate update, evaluate $\hat{\Delta}_k$, and roll back if $\hat{\Delta}_k > \tau$. Sweeping τ on the PPO Humanoid audit (Figure 2, right) shows a threshold flagging the worst-$\hat{\Delta}_k$ updates retains $\sim 80\%$ of the cumulative improvement while removing $\sim 60\%$ of the harmful updates by count. We validate this in actual training in Section 8, where an active rollback gate reduces the variance of training trajectories without degrading mean return. We present this strictly as a feasibility demonstration—evidence that the diagnostic has operational value—rather than a proposed control algorithm.

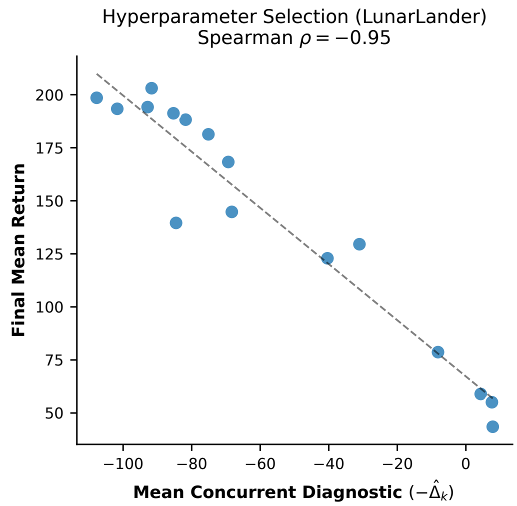

Figure 4: **Hyperparameter ranking via the diagnostic.** Mean concurrent $-\hat{\Delta}_k$ vs. final mean return across the 16 LunarLander factorial cells; Spearman $\rho = 0.95$. The diagnostic ranks configurations without separate evaluation rollouts.

Robustness and Sanity Checks. To ensure these findings are not artifacts of specific implementation details, we provide extensive robustness checks in the Appendix. We verify the start-state bias signal survives on fully independent held-out rollouts to rule out Monte Carlo leakage (Section D). We show the dominance holds across a 200-cell hyperparameter grid of value-epochs and clip thresholds (Figure 5), and that the failure of $\widehat{\epsilon}_u$ persists across alternative scalarizations like the sup-norm and 95th-percentile (Section E). Finally, we confirm the algebraic identity holds to machine precision in finite-state RandomMDPs (Section B) and that non-linear Random Forest combinations do not improve upon the additive identity.

8 Discussion

Our results expose a strict boundary in actor-critic optimization: because the critic is optimized on the rollout distribution, the start-state bias remains an intrinsically retrospective quantity.

What practitioners should do today: retrospective auditing. The immediate prescription is to log $\widehat{\hat{\Delta}}_k$ alongside $\widehat{\epsilon}_u$ in any actor–critic codebase. While the boundary prevents its use as an online loss term, practitioners can embrace its retrospective nature operationally:

(1) *Hyperparameter ranking.* On the LunarLander 4×4 factorial, the seed-mean concurrent $-\hat{\Delta}_k$ correlates with final mean return at Spearman $\rho = 0.95$ across the 16 cells (Figure 4). The diagnostic ranks configurations without expensive held-out evaluation rollouts.

(2) *Checkpoint-and-rollback gate.* As demonstrated in Section 7, practitioners can compute V_{k+1} on a candidate update and roll back parameters if $\hat{\Delta}_k > \tau$. In an actual (non-retrospective) training experiment on Humanoid-v5, a rollback gate rejected 38.3% of updates, tightened the cross-seed standard deviation by 23%, and matched vanilla PPO's mean return. We frame this strictly as a feasibility demonstration showing the operational value of the diagnostic, rather than a state-of-the-art control algorithm.

Implications for critic-objective design. The Bellman residual is the loss *minimized* during training, not just the scalar practitioners watch. Proposition 2 suggests a different target: drive down $\hat{\Delta}_k$. However, our ν-weighted regularizer experiment shows that a simple additive penalty on start states leaves the retrospective boundary intact. A successful objective may require much tighter Lipschitz-control of $|\hat{\Delta}_k| \leq L \cdot W_1(\nu, d^{\pi_k})$ via spectral normalisation [Miyato et al., 2018, Bjorck et al., 2021]. Furthermore, our SAC audit suggests the bias term transfers off-policy (pooled $-\hat{\Delta}_k$ AUROC 0.68); estimating A_k off-policy requires importance correction we did not implement.

Limitations. $\widehat{\hat{\Delta}}_k$ inherits the variance of $\widehat{J}(\pi_k)$ on environments with few complete episodes per rollout (HalfCheetah ~ 2, Ant ~ 3); the dominance over $\widehat{\epsilon}_u$ holds nevertheless. The lag-1 collapse means $\hat{\Delta}_k$ cannot be used as an online controller; it is a structural observation, not a proposed algorithm. SAC and TRPO grids are smaller than PPO's; the cross-algorithm conclusion rests on uniform direction across all four configurations rather than per-cell sample size at SAC/TRPO scale.

9 Conclusion

In actor–critic RL, the standard Bellman residual measures the wrong distribution, so it cannot reliably certify update health. Across PPO, SAC, and TRPO (237 seeds, eight environments), the Bellman residual pools to AUROC 0.46 as a harm predictor. The identity $\Delta J_k = A_k - \hat{\Delta}_k$ exposes the structural reality: the start-state bias is the operative quantity. It pools to 0.68 and outperforms $-\widehat{\epsilon}_u$ on $234/236$ paired seeds ($p < 10^{-40}$). Critic monitoring and design in deep RL must shift from absolute, rollout-based TD errors toward relative, start-state-aware quantities.

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A Hyperparameter factorials

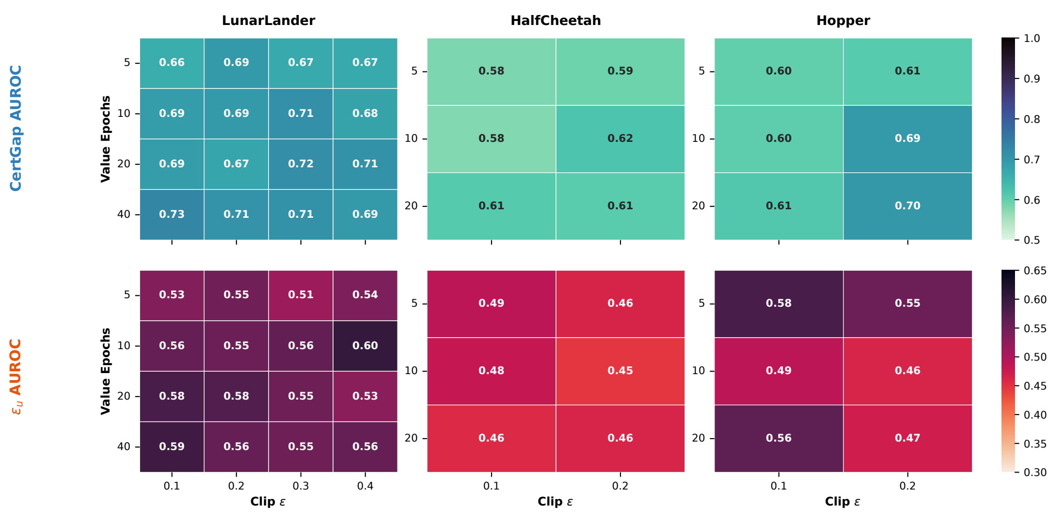

Figure 5: **Hyperparameter robustness** across LunarLander 4×4, HalfCheetah 3×2, and Hopper 3×2 factorials, 5 seeds per cell. *Top row*: $-\hat{\Delta}_k$ concurrent AUROC, dark (informative) on every cell. *Bottom row*: $-\widehat{\epsilon}_u$ concurrent AUROC, near-chance on every cell. The pattern of the main text is not specific to a single tuning of PPO.

The dominance of start-state bias is not specific to a single PPO configuration. Figure 5 reports per-cell concurrent AUROCs across three factorials: a 4×4 value-epochs $\times$ clip-ϵ grid on LunarLander, a 3×2 grid on HalfCheetah, and a 3×2 grid on Hopper, with an additional $10\times$ value-learning-rate sweep at the default cell of HalfCheetah and Hopper (the configuration that most directly varies critic convergence rate). Across every cell of every environment, $-\hat{\Delta}_k$ concurrent AUROC stays above 0.55, while $-\widehat{\epsilon}_u$ stays at or below 0.55. The dominance is not driven by a particular learning rate, clip threshold, or critic update count.

B Tabular verification of the identity

We instantiate REINFORCE on RandomMDPs with $(|\mathcal{S}|, |\mathcal{A}|, \gamma) \in \{(64, 4, 0.95), (128, 8, 0.90), (32, 2, 0.99)\}$. Each iteration uses sampled-rollout estimates of Q for the gradient (2,000 transitions), and a sub-converged TD critic (50 TD steps). We compute A_k via (4) and $\hat{\Delta}_k$ via the trained critic, with ΔJ_k obtained exactly via $\nu^\top (I - \gamma P^\pi)^{-1} r^\pi$. Figure 6: median residual $5{\times}10^{-16}$, max $4{\times}10^{-13}$. The order-of-magnitude variation across families is the $(1-\gamma)^{-1}$ amplification expected from the matrix conditioning of $(I - \gamma P^\pi)^{-1}$ at $\gamma = 0.99$.

C Proof of Proposition 2

Take $S = 1$, $A = 2$, $r = (0, 0)^\top$, $P(s \mid s, a) = 1$ for both a. Then $Q^\pi = \mathbf{0}$ for every π, so $J(\pi) = 0$. Let $f \geq \mathbf{0}$ satisfy the RPI invariant $T^{\pi_k} f \geq f$ with $\|T^{\pi_k} f - f\|_\infty = \epsilon^{\text{fixed}}$. Since

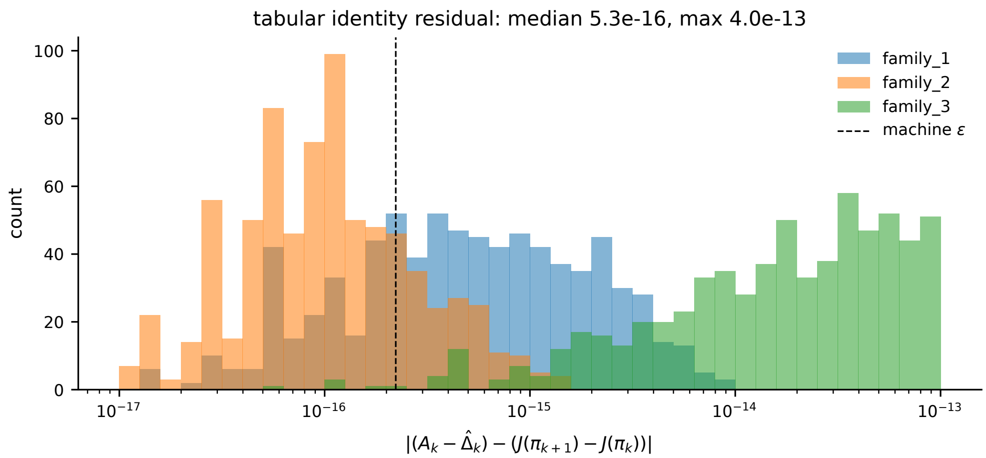

Figure 6: **The improvement identity (Proposition 1) holds to machine precision in tabular RandomMDPs.** Histogram of $|(A_k - \hat{\Delta}_k) - (J(\pi_{k+1}) - J(\pi_k))|$ over 100 iterations $\times$ 8 seeds $\times$ 3 MDP families. Median residual $5{\times}10^{-16}$, max $4{\times}10^{-13}$. Function approximation is not the cause.

$T^{\pi_k} f(a) = \gamma \bar{f}^{\pi_k}$ for both a where $\bar{f}^{\pi_k} = \sum_a \pi_k(a) f(a)$, let $f(a_1) > f(a_2)$ so $\bar{\pi}_k = a_1$. Then $A_k = \delta_k(f(a_1) - f(a_2))$, $C_k = 2\delta_k$, giving $A_k/C_k = (f(a_1) - f(a_2))/2$. The RPI condition at a_1 gives $\gamma \bar{f}^{\pi_k} \geq f(a_1)$, from which $f(a_1) - f(a_2) \leq \epsilon_u$. Therefore

$$\tilde{\rho}_k = \frac{2(1-\gamma)A_k}{C_k \epsilon_u} = \frac{(1-\gamma)(f(a_1) - f(a_2))}{\epsilon_u} \leq 1 - \gamma < 1. \qquad \square$$

D Estimator agreement (start-state vs. residual-based $\hat{\Delta}_k$)

The start-state estimator (6) and the alternative residual-based estimator $\widehat{\hat{\Delta}}_k^{\text{res}} = \mathbb{E}_{B_k}[r + \gamma V_{k+1}(s') - V_{k+1}(s)]/(1-\gamma)$ are different empirical proxies for the same population $\hat{\Delta}_k$. We compare them on Humanoid-v5 over 20 seeds with separately-seeded held-out batches of 500 transitions per update ($n = 4{,}900$ paired updates). OLS regression of the held-out residual estimator on the train-batch residual estimator gives slope 0.97, intercept +2.15, Pearson $r = 0.80$ (95% CI $[0.79, 0.81]$), $R^2 = 0.64$. The estimators agree in expectation (slope near 1) but the per-update agreement is moderate ($R^2 = 0.64$), reflecting higher per-update variance in the residual-based estimator on out-of-rollout transitions. We report the start-state form throughout because it is faithful to the formal definition (6); the alternative form recovers a higher pooled cert-gap AUROC by virtue of having $\sim 2{,}048$ samples per update versus $|\mathcal{S}_0|$ samples for the start-state form, but is closer to the rollout that just trained the critic.

E Robustness of ϵ_u across scalarizations

The headline ϵ_u in Table 1 uses mean-squared TD error post-critic-update (the value-function loss every PPO and SAC implementation we surveyed minimizes). Figure 7 reports the same quantity computed under two alternative scalarizations on Humanoid-v5: the sup-norm $\|T^\pi V - V\|_\infty$ (corresponding to the worst-case theoretical ϵ_u) and the 95th-percentile of $|T^\pi V - V|$ (a robust variant). All three sit at chance. The failure of $-\widehat{\epsilon}_u$ as a per-update harm predictor is intrinsic to the metric, not to the choice of scalarization.

F Per-seed paired comparison

Table 1 in the main text reports per-(algorithm, environment) median AUROC. Figure 8 visualises the seed-paired comparison for the PPO main grid.

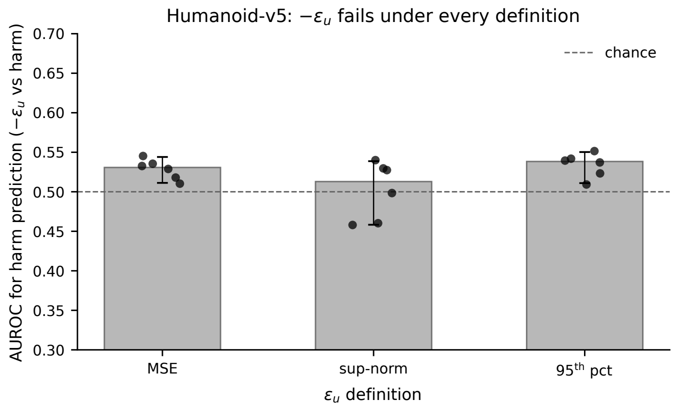

Figure 7: $-\widehat{\epsilon}_u$ **fails as a harm predictor under three definitions on Humanoid-v5.** Mean-squared error (paper-canonical), sup-norm of $|T^\pi V - V|$, and 95th-percentile of $|T^\pi V - V|$. Six seeds; all three definitions sit at or near chance.

G Experimental details

Environments and seed counts. PPO main: LunarLander-v3, CartPole-v1, Acrobot-v1 at 200k–300k env steps; Hopper-v5, HalfCheetah-v5, Walker2d-v5, Ant-v5 at 300k steps; Humanoid-v5 at 500k steps; 20 seeds per environment (160 runs). PPO-MSE ablation: same eight envs, 5 seeds each (40 runs). Hyperparameter factorials (PPO): LunarLander 4×4, 5 seeds/cell (80 runs); HalfCheetah 3×2 plus 3 value-lr cells, 5 seeds each (45 runs); Hopper 3×2 plus 3 value-lr cells, 5 seeds each (45 runs). ϵ_u-scalarization subset (Humanoid): 6 seeds with three Bellman-residual scalarizations logged. Estimator-agreement subset (Humanoid): 20 seeds with 500-transition held-out batch logging. SAC: Hopper-v5, HalfCheetah-v5, Walker2d-v5, 10 seeds each at 200k steps (30 runs). TRPO: LunarLander-v3, Hopper-v5, 10 seeds each at 150k steps (20 runs). Cross-algorithm total: ∼540 training runs, ∼86,000 logged updates. Environments via Gymnasium [Brockman et al., 2016].

Algorithm configurations. *PPO*: GAE [Schulman et al., 2015b] with γ=0.99, λ=0.95; value-epochs 10, policy-epochs 10, clip-$\epsilon = 0.2$; batch size 2048; minibatch 64; policy learning rate $3{\times}10^{-4}$, value learning rate 10^{-3}; gradient clipping at 1.0; no entropy bonus. (64, 64)-tanh MLPs with orthogonal initialization. Continuous-action environments use a Gaussian policy with state-independent log-standard-deviation. *PPO-MSE* replaces the clipped value loss with raw MSE; otherwise identical. *SAC*: standard twin-Q networks with target smoothing, automatic entropy temperature, replay buffer 10^6, batch size 256, learning rate $3{\times}10^{-4}$, $\gamma = 0.99$, $\tau = 0.005$. ***TRPO***: KL constraint $\delta = 0.01$, conjugate-gradient 10 iterations, line-search backtracks 10, policy step from natural gradient; same value network and GAE configuration as PPO. All implementations in PyTorch on CPU.

Diagnostic logging. Per update we log $\widehat{A}_k$, $\widehat{\hat{\Delta}}_k$ (start-state form), $\widehat{\epsilon}_u$ (mean squared TD error on rollout transitions, post-critic-update), and $\widehat{\Delta J}_k = \widehat{J}(\pi_{k+1}) - \widehat{J}(\pi_k)$ from successive rollout batches. The harmful label $1[\widehat{\Delta J}_k < 0]$ is undefined on the final update of each run. The same logging schema applies to PPO, PPO-MSE, SAC, and TRPO with appropriate definitions of “rollout batch” and the post-update critic.

Reproducibility. A single command (`make figures` after `pip install -e .`) regenerates every figure in this paper from the released update logs. The full pipeline (`make experiments`, equivalently `python scripts/run_all.py -workers 6 -skip-if-exists`) takes ~ 13 wall-clock hours on an 8-core CPU. Code, run logs, and `REPRODUCIBILITY.md` are released with the paper.

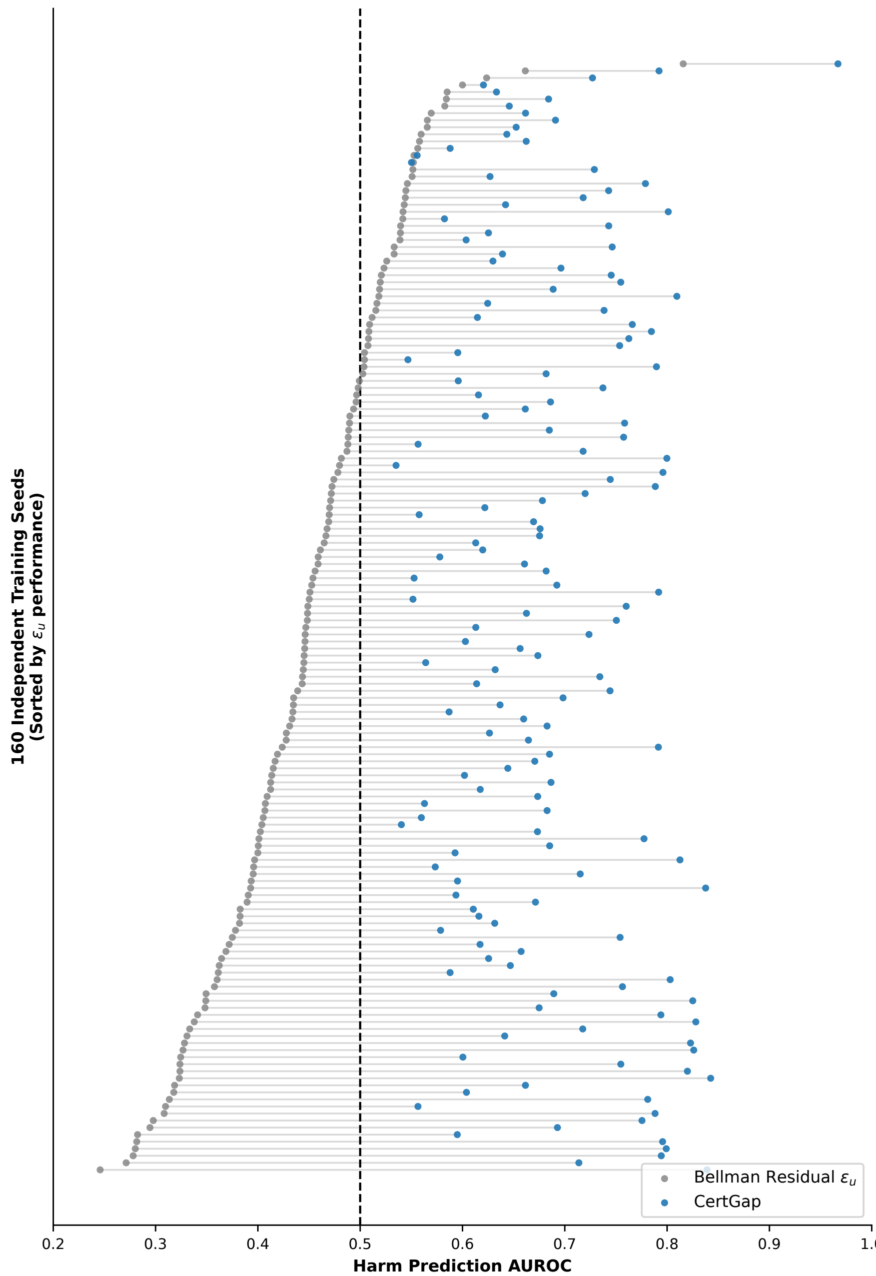

Figure 8: **Per-seed pairwise comparison.** Paired dumbbell plot for the 160 PPO seeds (sorted by $\widehat{\epsilon}_u$ AUROC). Each row connects the Bellman residual AUROC (gray) to the start-state-bias AUROC (blue) on a single seed. The bias term beats the residual on $158/158$ paired seeds (one tied), $p < 10^{-27}$.

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